



Medical Imaging & Big Data

Data Science Master's Degree

2024 – 2025

Deep Learning for Brain Tumor Segmentation: From Dropout to Residual U-Nets with Augmentation

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- Phase 1: Baseline U-Net with Dropout
- Phase 2: Data Augmentation Integration
- Phase 3: Residual Connections in U-Net
- Phase 4: Combined Enhancements
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Introduction and Data Preprocessing

This project focuses on the development of a deep learning pipeline for **semantic segmentation of brain tumors** from MRI scans.

The goal is to explore and progressively improve a U-Net-based architecture by incorporating key enhancements such as **dropout regularization**, **data augmentation**, and **residual connections**.

The data entailed is taken from the **BRATS** dataset, which includes pre-operative MRI scans from two groups:

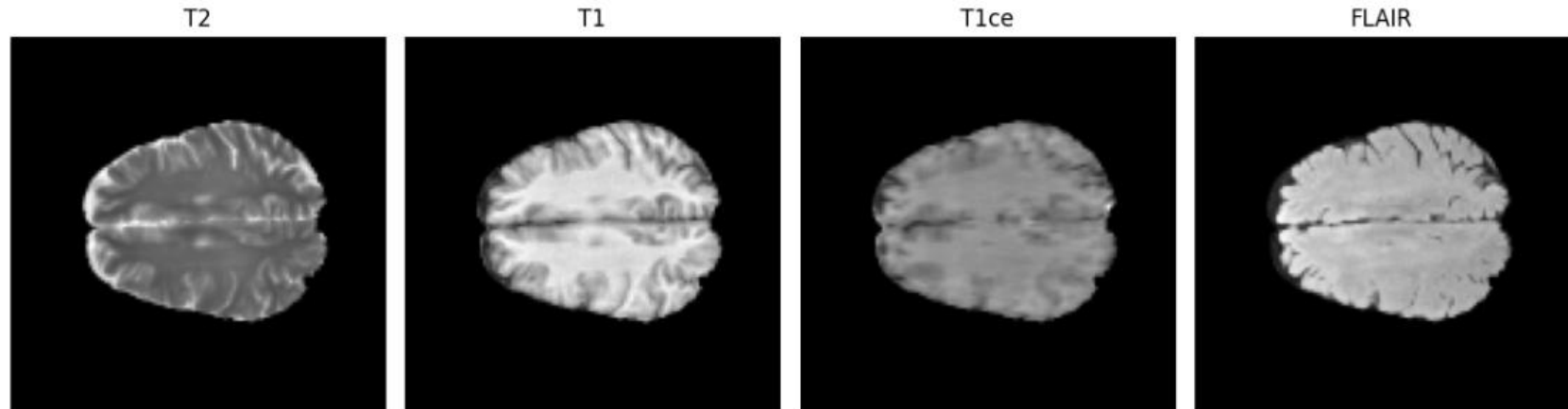
- LGG (Low-Grade Glioma)
- GBM (Glioblastoma Multiforme)

Introduction and Data Preprocessing

Each sample contains multimodal 3D MRI volumes, typically with **4 channels**:

- T1-weighted
- T1c (contrast-enhanced)
- T2-weighted
- FLAIR

MRI Modalities for Random Slice #434



Introduction and Data Preprocessing

To prepare the data for **modeling**:

- Volumes were resized to $120 \times 120 \times 77$ voxels to ensure uniform input shape
- All voxel intensities were scaled and normalized.
- Masks were converted to one-hot encoded arrays with binary channels for each class.

The **dataset** was split into:

- 67% for training □ 46 patients
- 33% for testing □ 24 patients

Split is stratified by patient volumes: each patient's 3D volume contributes 77 slices to the 2D training set. Preprocessed data is reshaped into 2D slices to support 2D convolution layers.

The aim is therefore to evaluate how architectural and training enhancements affect segmentation performance, measured through **Dice similarity coefficient** on tumor subregions

Phase 1: Baseline U-Net with Dropout

The first phase is meant to implement a **baseline segmentation model** using a classic **U-Net** architecture with **dropout regularization**. The goal is to establish reference performance using core MRI inputs.

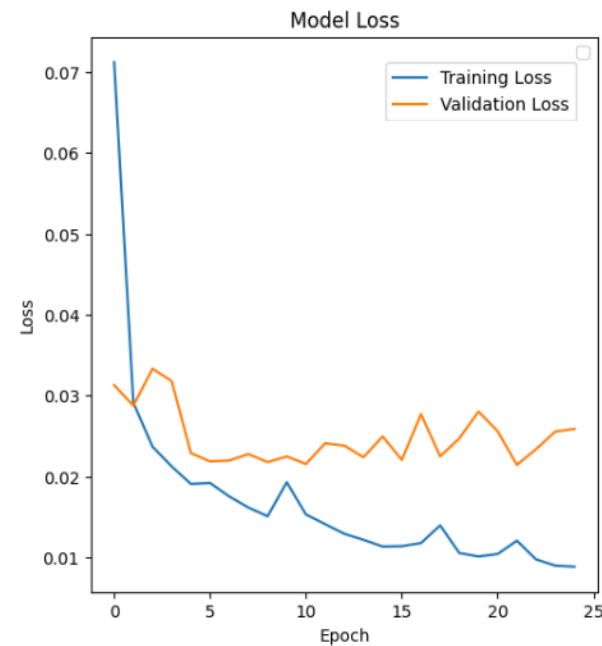
The model processes **2D MRI** slices of shape $120 \times 120 \times 4$ (four modalities per slice) and outputs a pixel-wise segmentation map with 4 **softmax-activated channels**.

The **architecture** was defined as follows:

- **Encoder:**
 - three encoder blocks, each with:
 - two Conv2D layers (ReLU, padding='same')
 - Dropout layers which rates increases with depth: $0.2 \rightarrow 0.2 \rightarrow 0.3$
 - MaxPooling2D to reduce spatial dimensions
- **Bridge:**
 - three Conv2D layers with 128 filters
 - final dropout with a higher rate of 0.4
- **Decoder:**
 - three decoder blocks, each composed of:
 - UpSampling2D for resolution
 - two Conv2D layers (ReLU, padding='same')
 - Dropout at each level: $0.3 \rightarrow 0.2 \rightarrow 0.2$
- **Output Layer:**
 - final Conv2D layer with 4
 - Softmax activation for multi-class segmentation per pixel

Phase 1: Baseline U-Net with Dropout

- The model was compiled and then fit using categorical loss and trained using:
- Batch size of 8
- 25 epochs



Phase 1: Baseline U-Net with Dropout

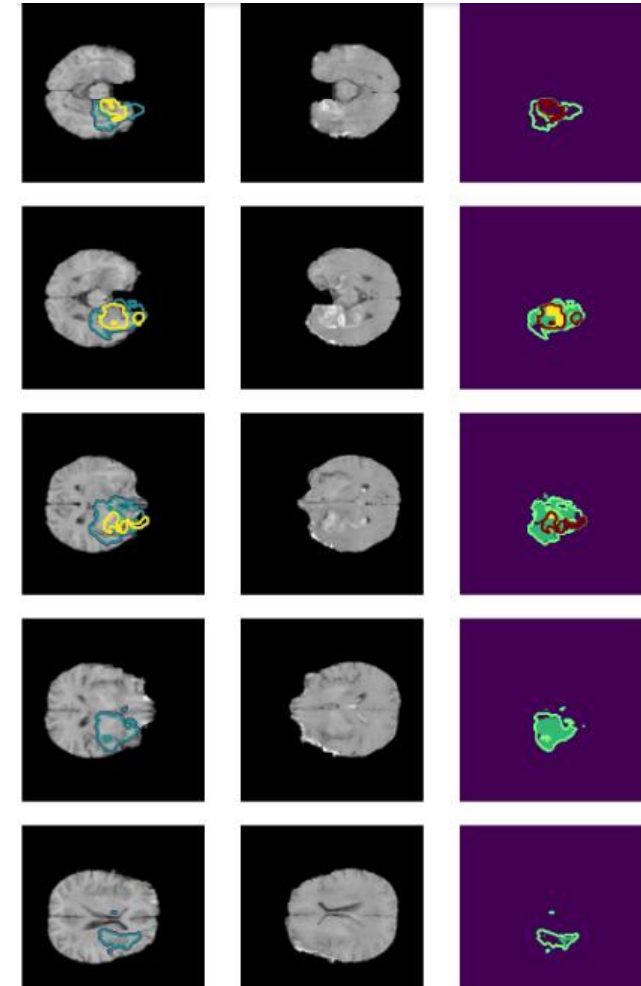
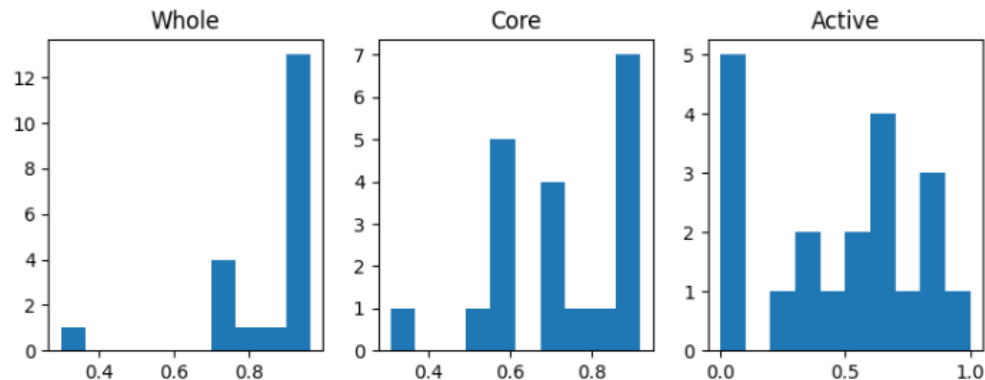
To assess segmentation quality, a **custom Dice coefficient function** was implemented to report that:

The Whole tumor segmentation is strong, with scores tightly clustered above 0.85. The mean is 84.4% and median is 89.9%, consistent with the best phases following.

The Core tumor performance is notably solid, with a mean of 74.2% and median of 79.7%, indicating improved consistency and reduced variance compared to the following improved phases.

The Active tumor Dice scores are significantly better than Phases 1 and 2, with a mean of 60.1% and median of 72.0%, and a clear concentration of values above 0.6.

Whole Dice: mean = 84.9% Median = 91.4%
Core Dice: mean = 72.3% Median = 73.2%
Active Dice: mean = 48.1% Median = 56.5%



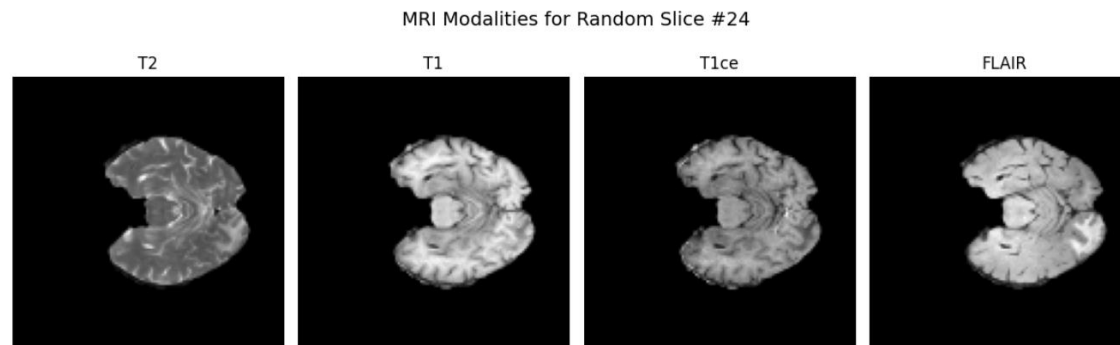
Phase 2: Data Augmentation Integration

In this phase, we aimed to improve generalization by increasing **data diversity**. We retained the U-Net architecture with dropout from Phase 1 and introduced a custom data augmentation pipeline for only 50% of the training data (i.e., 23 patients) applied directly to training slices.

A custom augmentation function was applied the training set, producing diverse transformed images:

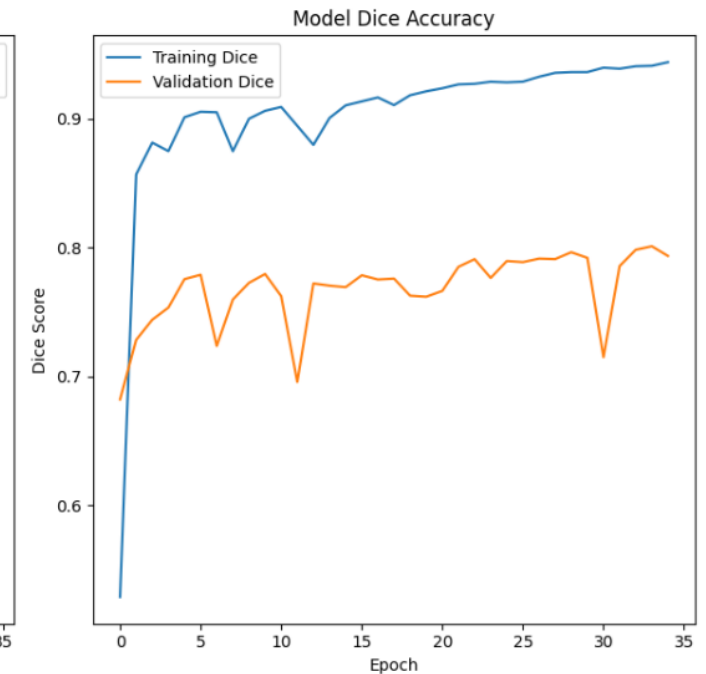
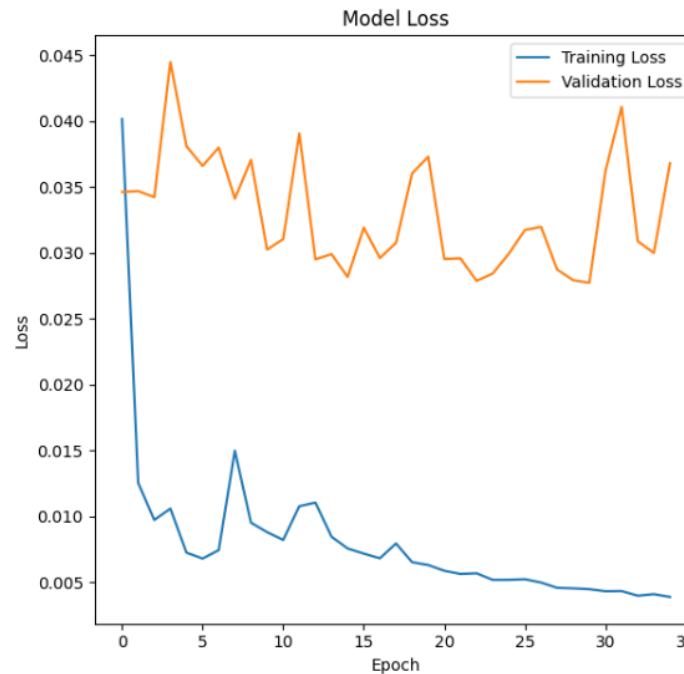
- Spatial: horizontal/vertical flips, random image roll
- Intensity: brightness, contrast jitter, multiplicative scaling

These augmentations aim to simulate realistic scanning variations and anatomical shifts.



Phase 2: Data Augmentation Integration

- In this case, the U-Net model previously defined was fit with:
 - batch size of 8
 - 35 epochs
- Also in this case, a custom Dice coefficient accuracy function was implemented.



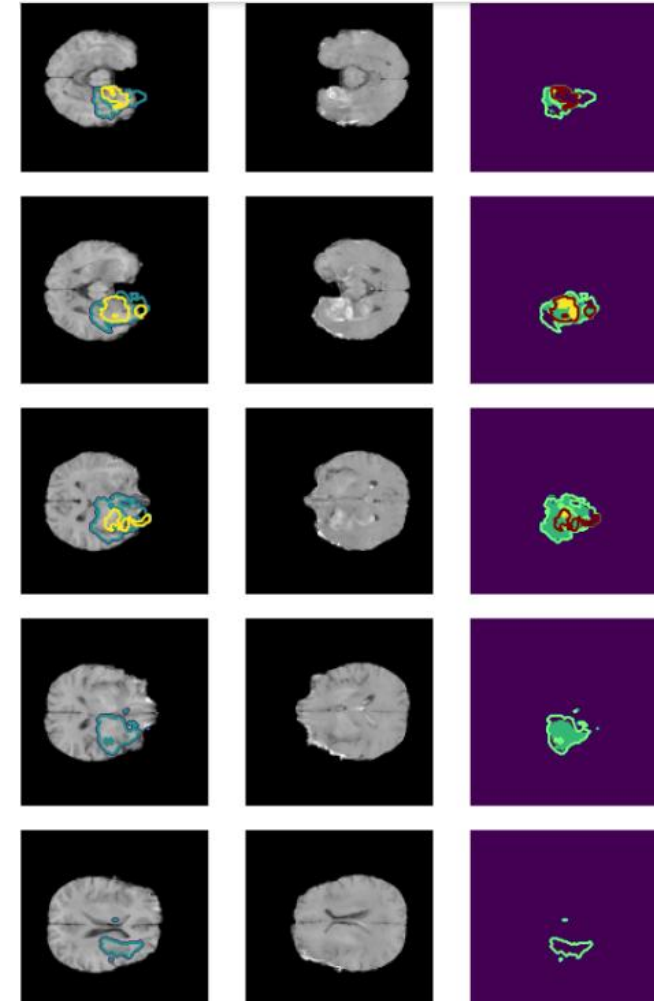
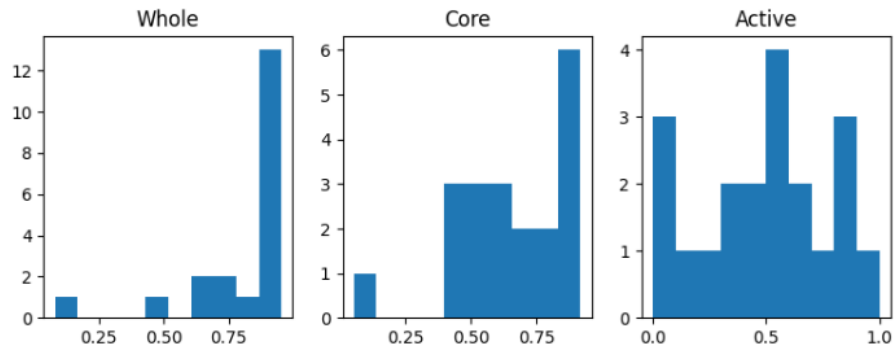
Phase 2: Data Augmentation Integration

The Whole tumor segmentation performs fairly well, with a mean of 79.9% and a high median of 89.1%, suggesting some low outliers but generally accurate results.

The Core tumor scores are more dispersed, with a mean of 65.4% and median of 68.4%, indicating less consistent segmentation across patients.

The Active tumor Dice scores are the weakest, with a mean of 49.1% and a median of 54.6%, showing a broad spread and multiple predictions under 0.5 — a sign of difficulty segmenting fine-grained tumor structures.

Whole Dice: mean = 79.9% Median = 89.1%
Core Dice: mean = 65.4% Median = 68.4%
Active Dice: mean = 49.1% Median = 54.6%



Phase 3: Residual Connections in U-Net

The aim of this phase is to explore architectural improvements to the U-Net from Phase 1 by integrating **residual connections** within the encoder, bridge and decoder blocks. The goal is to enhance the model's capacity to learn deeper, more expressive features without degradation in training performance.

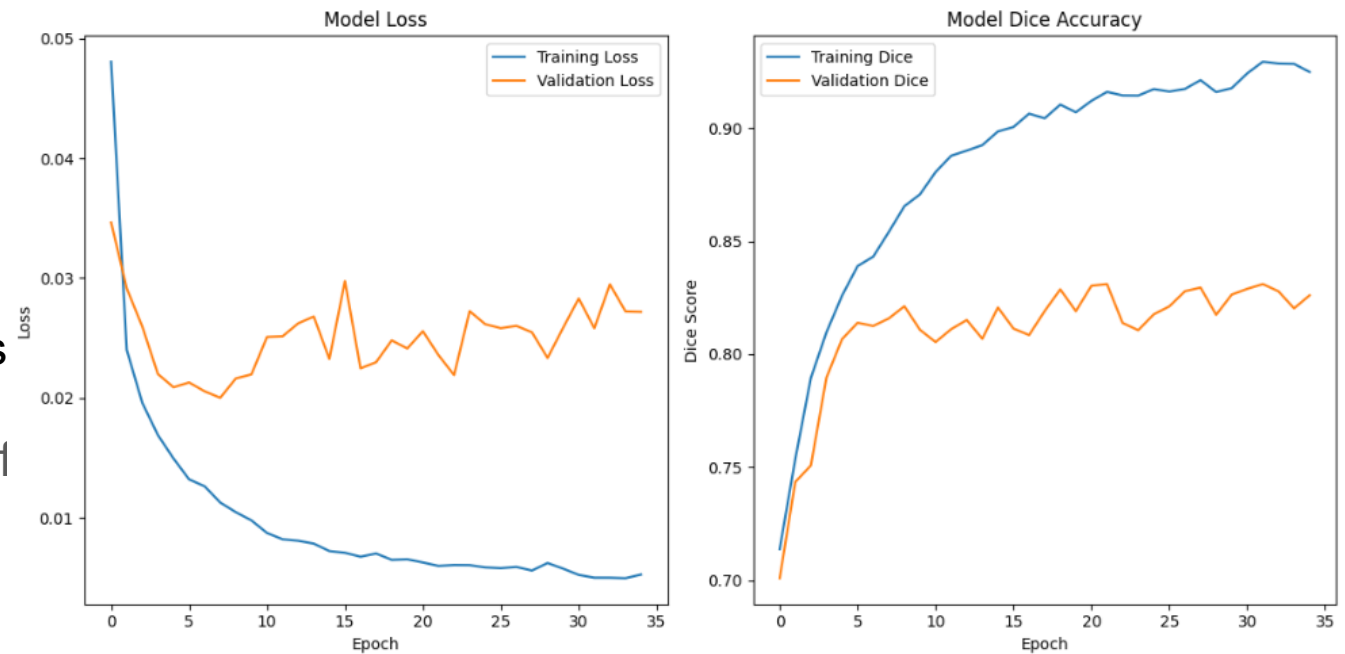
Each block includes:

- Two Conv2D layers (3×3)
- Dropout with increasing rate
- A 1×1 Conv2D shortcut to match dimens

In this case, the U-Net model previously de

- batch size of 8
- 35 epochs

Note: no data augmentation was employes in this phase



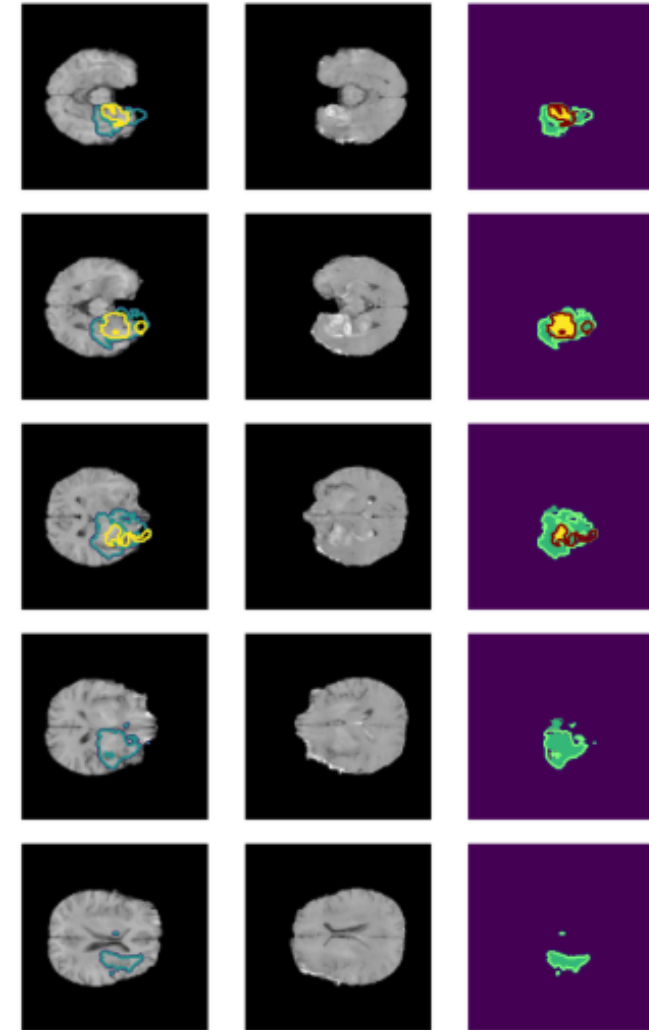
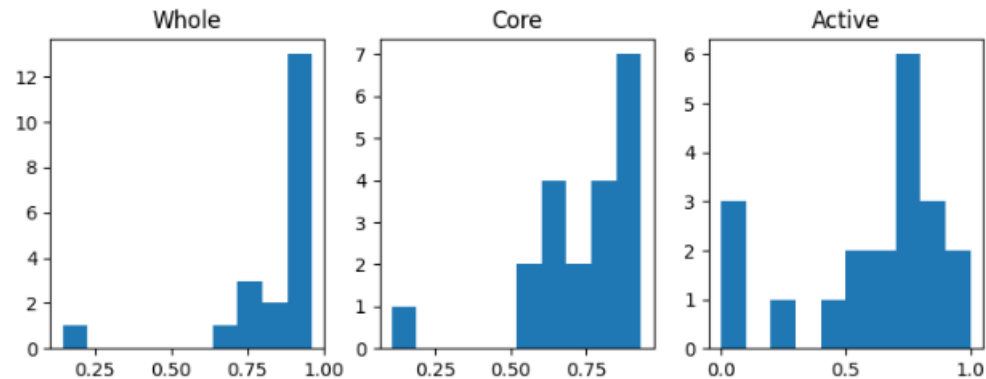
Phase 3: Residual Connections in U-Net

The Whole Dice scores are highly concentrated above 0.85, with a mean of 84.4% and median of 89.9%, indicating strong and consistent segmentation performance.

The Core tumor Dice scores show slightly more variability, with a mean of 74.2% and median of 79.7%, reflecting good but less uniform segmentation.

The Active tumor segmentation is the most challenging, with wider spread and a mean Dice of 60.1%. However, a median of 72.0% shows that many predictions were still accurate.

Whole Dice: mean = 84.4% Median = 89.9%
Core Dice: mean = 74.2% Median = 79.7%
Active Dice: mean = 60.1% Median = 72.0%



Phase 4: Combined Enhancements

The final experimental phase aims to **combine the strengths** of residual connections and data augmentation to try to **maximize model generalization** and segmentation precision.

The ***Augmentation*** function includes:

- Spatial transforms: roll, flip (horizontal & vertical)
- Intensity changes: scaling, brightness and contrast variation

The final training set contains data: original + augmented + shuffled split

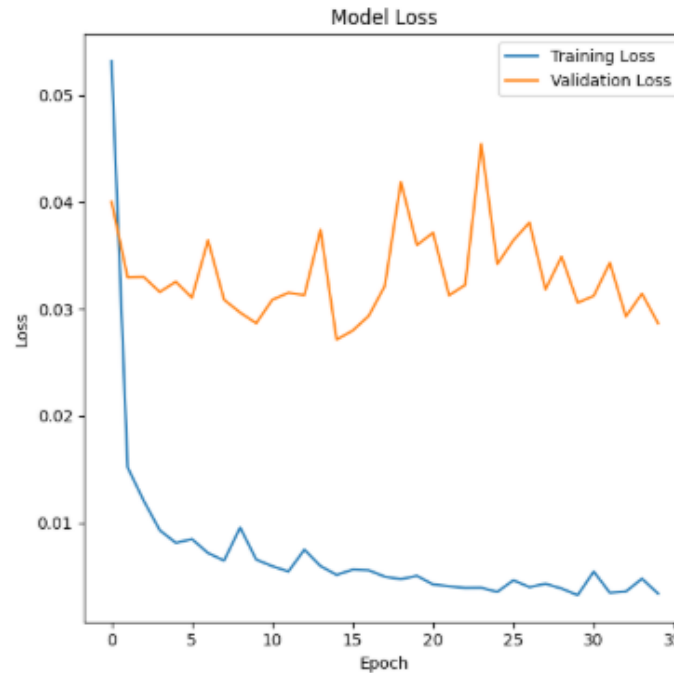
The is therefore a ***Residual U-Net***, with:

- encoder–decoder using residual blocks, where each block includes:
 - Two Conv2D layers (3×3) + ReLU
 - A 1×1 Conv2D shortcut path
 - increasing Dropout rates followed by ReLU
 - Output: Conv2D(4) with softmax activation for 4-class segmentation

Phase 4: Combined Enhancements

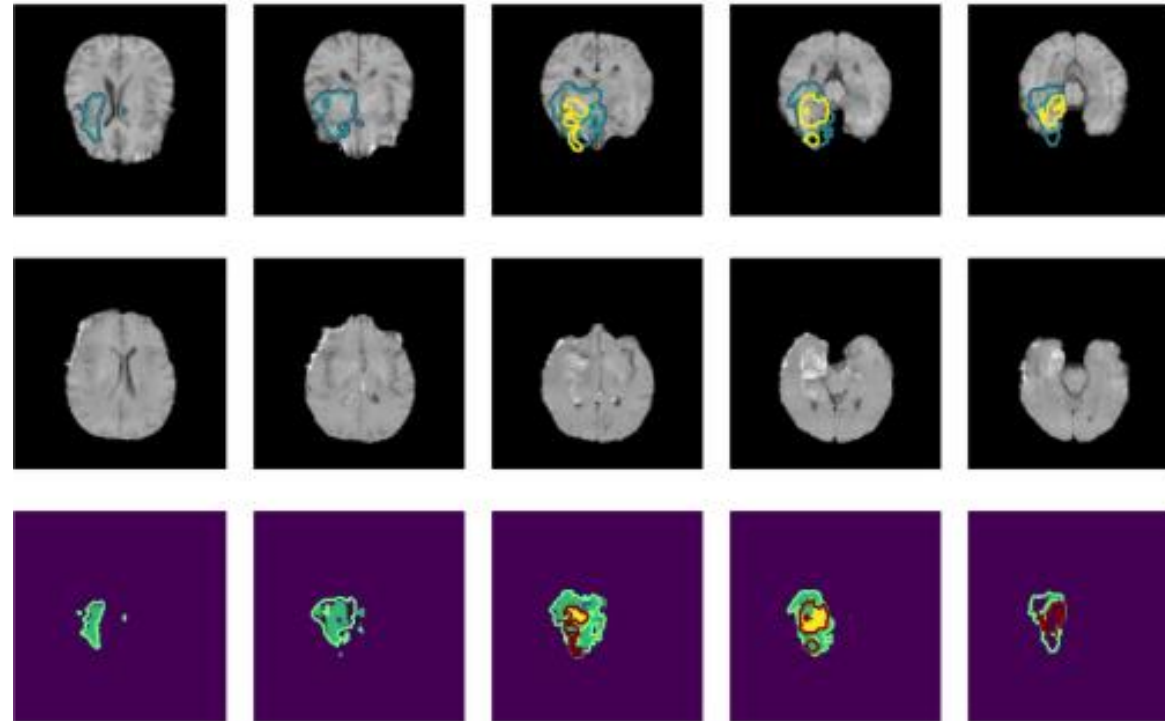
In this case, the U-Net model previously defined was fit with:

- batch size of 8
- 35 epochs



Phase 4: Combined Enhancements

In this phase, the segmentation masks appear to be the visually clear.



Results Comparison & Conclusios

Across all phases, we observed steady improvements in training behavior and segmentation quality:

- **Phase 1** showed solid Dice performance but also signs of overfitting and masks with irregular, fragmented boundaries.
- **Phase 2** slightly lower validation accuracy and noisy masks suggest overfitting to augmented data.
- **Phase 3** offered the best balance: with a validation Dice of **0.82** and noticeably cleaner, more coherent segmentations, especially in active tumor regions.
- In **Phase 4**, despite reaching the lowest training loss and highest Dice during training, validation Dice slightly dropped. However, the **segmentation masks** in Phase 4 were the most visually accurate, capturing finer details and tumor borders with greater precision than any prior model.

Phase	Train Loss	Validation Loss	Train Dice Accuracy	Validation Dice Accuracy
Phase 1	0.0087	0.0259	0.8755	0.8188
Phase 2	0.0041	0.0368	0.9435	0.7937
Phase 3	0.0052	0.0272	0.9265	0.8261
Phase 4	0.0035	0.0287	0.9454	0.7833

**THANK YOU FOR YOUR
ATTENTION!**